VIS2026

mcml
Munich Center for Machine Learning

Grand Unified Grammar of Graphics (GUGOG)

A workshop at IEEE VIS 2026

Cross-system comparison, meta-theoretical reflection and community consensus building for visualization grammars

WHEN

Monday 9 November 2026
1:00–4:30 PM EST

WHERE

Boston, Massachusetts

CO-ORGANIZED WITH

Munich Center for Machine Learning (MCML)

gugog-vis.github.io/2026

Half-day workshop

Programme

Monday 9 November 2026 · 1:00–4:30 PM EST

Session 1

1:00–2:30 PM

1:00 Opening

1:05 **Balancing constriction and expansion for domain-specific grammars**
Jason Cory Brunson

1:15 **The Grammar of Graphics Needs Verbs: Towards a GUGOG**
Josh M. Pollock

1:25 **Semantic Validity vs. Graphical Arrangement in Visualization Grammars**
Cynthia A. Huang & Matthew Kay

1:35 **Panel: Towards a Universal Visualisation Language**
Andrew McNutt · Eugene Wu · Josh Pollock

Session 2

3:00–4:30 PM

3:00 **Can the Grammar of Graphics Describe Data Glyphs?**
Barbara Nascimento, João M. Cunha & Evgheni Polisciuc (pre-recorded)

3:10 **On the Inconsistency in Grammar Evaluations**
Lemara Williams

3:20 **Mind Your vs and Qs: A Categorical Framework for Specifying Visualization Components**
Hannah Aizenman, Mikael Vejdemo-Johansson, Thomas Caswell & Michael Grossberg

3:30 Discussion group formation

3:35 Discussion groups

4:10 Report back, synthesis, call to action

4:25 Closing

Coffee break 2:30–3:00 PM

Acknowledgements

ORGANIZERS

Cynthia Huang
LMU Munich / MCML

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PROGRAM COMMITTEE

Cynthia Huang (chair)
LMU Munich / MCML

Harriet Mason
University of Queensland

Sherry Zhang
University of Texas at Austin

HOSTED BY



Thank you to the **IEEE VIS 2026 Organizing Committee** for hosting GUGOG, and to all authors, panelists and participants.

CO-ORGANIZED WITH



GUGOG is co-organized with and supported by the **Munich Center for Machine Learning (MCML)**, the joint AI research center of LMU Munich and TU Munich and one of six national AI Competence Centers in Germany, funded by the German Federal Ministry of Research, Technology and Space and the Free State of Bavaria.

Munich Center for Machine Learning

MUNICH CENTER FOR MACHINE LEARNING

Co-organizer of GUGOG @ IEEE VIS 2026

LMU LUDWIG-MAXIMILIANS-UNIVERSITÄT MÜNCHEN

Technische Universität München

Bundesministerium für Forschung, Technologie und Raumfahrt

HIGHTECH Agenda Bayern

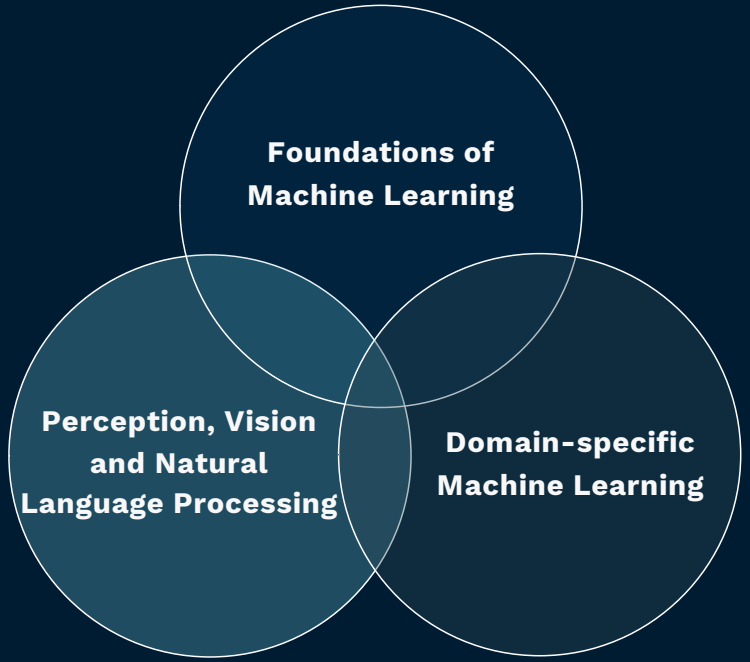
Bayerisches Staatsministerium für Wissenschaft und Kunst

GET TO KNOW THE MCML

- Joint AI research center of LMU & TUM
- One of six national AI Competence Centers in Germany
- Funded by the German Federal Ministry of Research, Technology and Space & Free State of Bavaria
- 11 research areas in the fields of application-oriented ML and foundational research
- Developing responsible and trustworthy AI
- Supporting the next generation of AI researchers



MCML MISSION



3 RESEARCH AREAS

- **VISION**
  Advancing machine learning methods for application-oriented fields such as medicine, biology, and ethics
- **RESEARCH**
  Interdisciplinary research at the interface of computer science and cognitive science
- **OUTREACH**
  Educational programs and outreach activities to promote machine learning and AI knowledge

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PANEL

Towards a Universal Visualisation Language

Is a universal visualisation language a coherent goal, or does the diversity of grammars reflect genuine differences in what people need to express?



Andrew McNutt

University of Utah (incoming UBC)

PROGRAMMING INTERFACES & HCI

Builds tools and theories that turn domain expertise into software that is approachable, trustworthy, and respectful of user agency.



Eugene Wu

Columbia University

DATABASES & SYSTEMS

Co-director of the Data, Agents, and Processes Lab (DAPLab). Research spans visualization, HCI, and core data systems.



Josh Pollock

MIT VIS group (recent PhD)

PL FOR VISUALISATION GRAMMARS

Applies programming language ideas to visualization grammars, looking for unifying abstractions. Creator of GoFish and Bluefish.

Moderated by **Cynthia Huang**, LMU Munich / MCML

DISCUSSION GROUPS

Which theme do you want to work on?

Pick a panel theme, or add your own, on the Excalidraw board.

1

Does universality make sense?

A shared core, or real differences in what people need to express?

2

What do we want from other communities?

Statistical graphics, HCI, diagrams, cartography, accessibility

3

Research or engineering?

Missing abstractions, shared modules, standards and tooling

4

5

QR code

Add ideas on Excalidraw